

② 12.1

$$S_{xx} = 7918.169 \quad S_{xy} = 7917.130$$

$$b_0 = \bar{y} - b_1 \bar{x} = 1.38257$$

$$\hat{y} = 1.38257 + 0.97461x$$

b.  $SS_T = 1567.800 \quad \Sigma(y - \bar{y})^2$   
 $SS_R = 1521.195 \quad \Sigma(\hat{y})^2$   
 $SS_E = 46.605 \quad \Sigma(y - \hat{y})^2$

$$R^2 = \frac{SSR}{SST} = \frac{7521.195}{9567.800} = 0.9938$$

⇒ reasonable

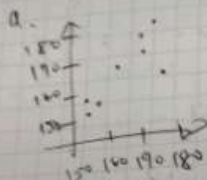
reject  $H_0$

idence that  
is not

差矣  $\Rightarrow$  blocking 有效

著不同

22



b.  $y = x \rightarrow \text{斜率} = 1$

$$b_1 = \frac{S_{xy}}{S_{xx}} = 0.82394$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$b_0 = 166.395 - (0.82394)(164) = 29.60035$$

$$\hat{y} = 29.60035 + 0.82394x$$

d.  $\hat{y} = 29.60035 + 0.82394x$

$$x = 159 \rightarrow \hat{y}$$

$$\Rightarrow \hat{y} = 158.958$$

c.  $n = 5$

$$\bar{x} = 166.000$$

$$\bar{y} = 166.395$$

$$S_{xx} = 568.000$$

$$S_{xy} = 468.000$$

23

10

a.  $x = \text{Income}$

$y = \text{BMI}$

b.  $n = 6$

$$\bar{x} = 40.0833$$

$$\bar{y} = 27$$

$$S_{xx} = 2730.283$$

$$S_{xy} = -402.85$$

$$b_1 = \frac{S_{xy}}{S_{xx}} = \frac{-402.85}{2730.283} = -0.14755$$

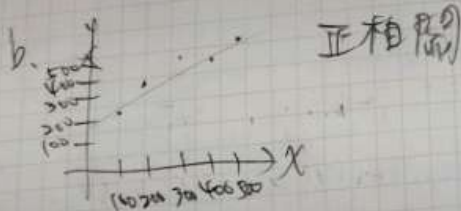
$$b_0 = \bar{y} - b_1 \bar{x} = 27 - (-0.14755)(40.0833) = 32.9144$$

$$\hat{y} = 32.9144 - 0.14755x$$

c. ANOVA

|                                                          | SS     | df | MS     | F      |
|----------------------------------------------------------|--------|----|--------|--------|
| Regression                                               | 59.442 | 1  | 59.442 | 18.609 |
| Error                                                    | 12.220 | 4  | 3.055  |        |
| Total                                                    | 71.662 | 5  |        |        |
| $R^2 = \frac{SSR}{SST} = \frac{59.442}{71.662} = 0.8291$ |        |    |        |        |

① a.  $n=5$   $S_{xx}=96080$   
 $\bar{x}=98$   $S_{xy}=64386$   
 $\bar{y}=395.6$   
 $b_1 = \frac{S_{xy}}{S_{xx}} = \frac{64386}{96080} = 0.67013$   
 $b_0 = \bar{y} - b_1 \bar{x} = 395.6 - 0.67013 \times 98$   
 $= 195.903$   
 $\Rightarrow \hat{y} = 195.903 + 0.67013x$

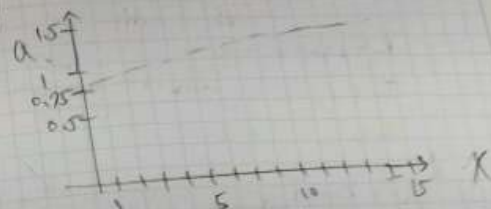


c.

|            | df | SS       | MS       | F      |
|------------|----|----------|----------|--------|
| Regression | 1  | 43146.93 | 43146.93 | 69.582 |
| Error      | 3  | 1860.27  | 620.09   |        |
| Total      | 4  | 45007.2  |          |        |

12.3

①



b.  $S_{xx} = \sum (x - \bar{x})^2 = 340$   $\bar{x} = 9.5$   
 $S_{xy} = \sum (x - \bar{x})(y - \bar{y}) = 1.85$   $\bar{y} = 1.04125$   
 $b_1 = 0.02309$   
 $b_0 = \bar{y} - b_1 \bar{x} = 1.04125 - (0.02309)(9.5)$   
 $= 0.86809$   
 $\hat{y} = 0.02309x + 0.86809$

$H_0: B_1 = 0$

$H_a: B_1 \neq 0$

$T = 2.23$   $p\text{-value} = 0$   $\alpha = 0.05$   
 reject  $H_0$

### C. ANOVA

|            | SS       | df | MS       | F       |
|------------|----------|----|----------|---------|
| Regression | 0.181243 | 1  | 0.181243 | 494.393 |
| Error      | 0.005132 | 14 | 0.000367 |         |
| Total      | 0.186375 | 15 |          |         |

$$\alpha = 0.05 \Rightarrow F_{0.05, 1, 14} = 4.60$$

$\Rightarrow$  reject  $H_0$

d.  $r^2$

$$r^2 = \frac{SSR}{SST} = \frac{0.181243}{0.186375} = 0.9725$$

$\Rightarrow$  very effective

(10)

x. Excel coefficient

$$\hat{y} = 93.619 - 1.251x$$

$\uparrow$                        $\uparrow$   
 $b_0$                        $b_1$

$$b. H_0: \beta_1 = 0$$

$$H_a: \beta_1 \neq 0$$

$$t = -10.471$$

$$df = n - 2 = 5$$

$$\alpha = 0.05 \Rightarrow t_{0.025, 5} = 2.571$$

$$|t| = 10.47 > 2.571 \Rightarrow \text{reject } H_0$$

$$c. b \pm t_{\alpha/2} \text{ of } SE(b_i) \quad -1.251 \pm 2.571(0.120)$$

$$b_1 = -1.251 \quad \Rightarrow (-1.559, -0.943)$$

$$SE(b_i) = 0.120$$

$$t_{0.025, 5} = 2.571$$

95% C.I. 落在

$$d. R^2 = 0.9564$$

$\rightarrow$  very effective